

V1 REQUIREMENTS,

TRACEABILITY, AND RISK BASELINE

Autonomous Semantic Navigation Robot

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Isaac Sim + ROS 2 Jazzy + Nav2

Version 0.9 – Development baseline draft – 24 September 2026

Purpose. Convert the V1 ConOps into measurable requirements, planned verification, configuration rules, and a lightweight risk baseline sufficient to begin implementation without creating unnecessary documentation overhead. Controlled status: suitable for implementation planning. Provisional targets must be confirmed by integrated benchmarking

and revised before the formal acceptance campaign.

DOCUMENT CONTROL		Lean development baseline
Field	Value	
Document owner	Jan Cuesta	
Project	Autonomous Semantic Navigation Robot	
Document identifier	ASN-V1-REQ-RISK	
Revision	0.9	
Status	Development baseline draft; provisional values remain controlled targets	
Authoritative content	This PDF/LaTeX baseline and its generated requirement, traceability, and risk CSV registers	
Related baseline	V1 Concept of Operations and Scope Baseline, Version 0.9	
Tailoring principle	Record only what supports implementation, verification, risk control, reproducibility, or release decisions	

Revision and approval

Version	Date	Change	Status
0.9	24 Sep 2026	Initial consolidated requirements, verification allocations, risk register, acceptance vocabulary, and budgets.	Review
1.0	Pending	Confirm provisional targets after integrated measurements and incorporate applicable confirmed robot constraints.	Frozen before acceptance

Role	Name / approval	Date
Document owner / developer	Jan Cuesta: _____	_____
Stakeholder / supervisor	_____	_____

Implementation gate. Version 0.9 is intentionally sufficient to begin architecture and implementation. Detailed test procedures are created when their feature is implemented. Before formal acceptance, every provisional target must be confirmed or revised through controlled evidence and this document must be issued as Version 1.0.

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01 / PURPOSE AND TAILORING

Enough control to build and defend the result

1. Purpose and scope

This document establishes the measurable V1 obligations derived from the ConOps. It covers simulation behavior, mapping, localization, navigation, live obstacles, recovery, stopping, performance, diagnostics, evidence, configuration control, acceptance, and the physical-robot transition boundary.

It does not contain node/topic/QoS details, implementation algorithms, complete test procedures, or physical-robot interface facts. Those belong to Steps 3–6 and to feature-level implementation records. This keeps the project focused on acquiring practical ROS 2, C++, SLAM, Nav2, integration, debugging, and validation experience.

1.1 Status meanings

Status	Meaning
Baseline candidate	Agreed V1 obligation; expected to remain unless implementation evidence reveals a conflict.
Provisional target	Initial measurable target; implementation may revise it only through recorded evidence and impact review before acceptance.
Monitoring only	Recorded for engineering insight but not an independent pass/fail condition.
Stretch	Valuable optional evidence that cannot block or compensate for the mandatory V1 release.

02 / ACCEPTANCE VOCABULARY

One interpretation for every trial

2. Controlled definitions

Term	Definition
Successful nominal trial	A valid goal is accepted; response and feedback limits are met; required system health remains acceptable; the goal is reached within timeout and final pose tolerances; the stopped condition is held for five seconds; no collision or unauthorized intervention occurs; and the required evidence is complete.
Failed trial	One or more mandatory success conditions are not met, including collision, timeout, excessive final error, manual steering, unsafe or incorrect terminal behavior, missing mandatory health, or system crash.
Invalid trial	The wrong or unidentified configuration, world, map, start, goal, procedure, or evidence set was used. It is retained, excluded from the pass-rate denominator, and repeated correctly.
Collision	Any registered contact between robot collision geometry and a wall, obstacle, desk, pillar, temporary box, or controlled moving object. Normal wheel-floor contact is excluded.
Manual intervention	Operator steering, repositioning, goal change, parameter change, reset, or unplanned simulator manipulation after the timed trial begins. Predefined scenario insertion/removal is excluded.
Automatic recovery	A finite configured action performed without operator input. Successful recovery does not make the trial fail; it is recorded as Pass - recovery used when all other criteria pass.
Ready	Stationary state in which mandatory processes, sensing, odometry, TF, localization, command ownership, logging, and stop controls satisfy their readiness checks.
Manual Safety Stop	Prioritized software stop that cancels motion and latches motion inhibition until explicit reset and readiness checks. It is not a certified physical emergency stop.
Simulated Emergency Stop	Highest-priority simulated motion-inhibition input. It bypasses normal mission sequencing, latches the inhibited state, and keeps diagnostics/evidence alive when practical. It does not validate physical safety.

Term	Definition
Blocking failure	Collision, uncontrolled motion, failure to stop, corrupted mandatory evidence, or another safety-related fault that prevents release until investigated and dispositioned.

3. Successful-trial decision rule

An automatic recovery is permitted. A nominal trial containing a safety-critical alarm, unresolved functional error, stale mandatory data, or unapproved recurring warning does not pass merely because the robot eventually reaches the goal. Logger severity alone is not the decision criterion; the project diagnostic classification is.

03 / PERFORMANCE AND TIMING BUDGET

Targets to confirm during integration

4. Current budgets

Area	Current value	Control
Cold startup to Ready	120 s	Provisional
Localization after initial pose	30 s	Provisional
Goal acknowledgment / first feed-back / first motion	2 s / 1 s / 5 s	Baseline candidate
Trial timeout	180 s	Baseline candidate
Final error	0.20 m and 10 degrees	Baseline candidate
Final stationary confirmation	5 s at $v \leq 0.01$ m/s and $ \omega \leq 0.02$ rad/s	Baseline candidate
Mapping speed	0.20 m/s, 0.60 rad/s	Baseline candidate
Navigation/recovery speed	0.25 m/s, 0.75 rad/s	Baseline candidate
Minimum obstacle separation	0.20 m	Provisional
Command watchdog	0.50 s stale; zero output within 0.10 s; stopped within 1.0 s	Provisional except stale period
LiDAR / odometry / dynamic TF	9 Hz / 27 Hz / 27 Hz minimum	Provisional
Critical data age / command latency	0.25 s / 100 ms maximum	Provisional
Physics / real-time factor	60 Hz; mean RTF ≥ 0.95	Provisional
Map resolution	0.05 m/cell	Provisional
Host temperature	Record CPU/GPU temperatures and throttling	Monitoring only

Temperature interpretation. Temperature alone does not fail robot software. A trial fails when thermal behavior causes a violated functional or timing requirement. Temperature and resource data are retained to select a sustainable launcher configuration.

04 / ATOMIC REQUIREMENT REGISTER

Measurable obligations

5. Register structure

The table below is the controlled human-readable register. The companion CSV is intended for sorting and implementation updates. Verification results remain Open until the corresponding software exists.

5.1 Operations

ID	Shall statement	Status	Verify
OPS-001	The system shall inhibit motion until the required processes, simulation clock, LiDAR, odometry, TF, localization, command path, logging, and stop controls satisfy the Ready checks.	Baseline candidate	Test
OPS-002	The controlled launcher shall reach Ready within 120 seconds of invocation.	Provisional target	Test
OPS-003	The system shall permit manual velocity control only in Mapping mode and shall prohibit manual steering during navigation acceptance trials.	Baseline candidate	Test
OPS-004	The system shall maintain only one authorized velocity-command source in each operating mode.	Baseline candidate	Test and inspection
OPS-005	After a normal goal cancellation, the system shall stop, report Canceled, return to Ready, and not resume the canceled goal automatically.	Baseline candidate	Test
OPS-006	Orderly shutdown shall cancel active motion, verify the stopped condition, preserve required evidence, avoid modifying the approved map, and terminate controlled processes in the documented order.	Baseline candidate	Test

5.2 Mapping

ID	Shall statement	Status	Verify
MAP-001	The mapping function shall construct the occupancy map from simulated LiDAR and wheel-derived odometry without using simulator ground truth as a mapping or navigation input.	Baseline candidate	Test and inspection
MAP-002	Mapping commands shall not exceed 0.20 m/s linear speed or 0.60 rad/s angular speed.	Baseline candidate	Test
MAP-003	The approved static occupancy map shall represent both rooms, both doorways, the corridor, turn pocket, permanent obstacles, and localization features without unexplained discontinuities or duplicated geometry.	Baseline candidate	Inspection and analysis
MAP-004	The initial occupancy-map resolution shall be 0.05 m/cell.	Provisional target	Inspection
MAP-005	The saved map and metadata shall identify the world, robot, SLAM configuration, and repository revision used to create them.	Baseline candidate	Inspection
MAP-006	The approved map shall load successfully after a clean restart.	Baseline candidate	Test
MAP-007	Navigation operation shall not overwrite or silently modify the approved map.	Baseline candidate	Test and inspection
MAP-008	After rigid 2D alignment, mapped doorway, corridor, and turn-pocket clear dimensions shall differ from programmed dimensions by no more than 0.10 m.	Provisional target	Analysis

ID	Shall statement	Status	Verify
MAP-009	After rigid 2D alignment, at least 95 percent of sampled permanent boundaries shall lie within 0.15 m of their programmed boundaries, and no included sample shall exceed 0.20 m error.	Provisional target	Analysis
MAP-010	The approved map shall contain neither a false occupied region that blocks a frozen nominal route nor a false free opening through a wall that the planner can treat as traversable.	Baseline candidate	Analysis and demonstration

5.3 Localization

ID	Shall statement	Status	Verify
LOC-001	Navigation startup shall load the approved map and accept an approximate initial pose for AMCL.	Baseline candidate	Test
LOC-002	The system shall achieve localization readiness from each of S1, S2, and S3 within 30 seconds after receiving a valid initial pose.	Provisional target	Test
LOC-003	Nominal localization readiness shall not require manual or autonomous initialization motion.	Baseline candidate	Test
LOC-004	Localization shall not report Ready unless map-to-odom is available, required sensor and TF data are fresh, pose age is no greater than 0.50 seconds, and estimated horizontal and yaw standard deviations remain no greater than 0.20 m and 10 degrees for three continuous seconds.	Provisional target	Test and analysis
LOC-005	The mission manager shall reject goals while localization is absent, stale, or outside its readiness limits.	Baseline candidate	Test
LOC-006	Loss or implausibility of localization during navigation shall stop mission motion and require restored readiness before further movement.	Baseline candidate	Fault-injection test
LOC-007	Simulator ground truth shall be used only for debugging and independent evaluation, not for mapping, localization, planning, or control.	Baseline candidate	Inspection and test

5.4 Navigation

ID	Shall statement	Status	Verify
NAV-001	The mission manager shall reject malformed, out-of-map, occupied, or operationally prohibited goals without commanding motion.	Baseline candidate	Test
NAV-002	The system shall acknowledge or reject a submitted goal within two seconds.	Baseline candidate	Test
NAV-003	The system shall provide the first navigation feedback within one second after goal acceptance.	Baseline candidate	Test
NAV-004	A valid accepted goal requiring movement shall produce the first nonzero navigation command within five seconds.	Baseline candidate	Test
NAV-005	While navigation is active, mission feedback shall update at least once per second with no unexplained gap longer than two seconds.	Baseline candidate	Test

ID	Shall statement	Status	Verify
NAV-006	Mission feedback shall identify mission state, elapsed time, remaining distance when available, recovery activity, required-system health, and warning or fault conditions.	Baseline candidate	Test and inspection
NAV-007	The system shall use the approved saved map for global planning and current LiDAR observations for local obstacle response.	Baseline candidate	Test and demonstration
NAV-008	Navigation and automatic-recovery commands shall not exceed 0.25 m/s linear speed or 0.75 rad/s angular speed.	Baseline candidate	Test
NAV-009	Successful goal completion shall have ground-truth position error no greater than 0.20 m.	Baseline candidate	Test and analysis
NAV-010	Successful goal completion shall have ground-truth heading error no greater than 10 degrees.	Baseline candidate	Test and analysis
NAV-011	Successful goal completion shall maintain the position and heading tolerances, linear speed no greater than 0.01 m/s, and angular speed no greater than 0.02 rad/s for five continuous seconds.	Baseline candidate	Test and analysis
NAV-012	A navigation trial shall reach a valid terminal state within 180 seconds after goal acceptance.	Baseline candidate	Test
NAV-013	In TB-A, the system shall detect the temporary box, use an available safe route, and reach the goal without contact or manual steering.	Baseline candidate	Test
NAV-014	In TB-B, the system shall detect the blocked route, exhaust only permitted recovery actions, stop, report failure, and return stationary to Ready.	Baseline candidate	Test
NAV-015	In TB-C, the system shall clear the removed temporary obstacle and continue the still-active goal when the recovery period has not expired.	Provisional target	Test
NAV-016	If the TB-C goal has already terminated, the system shall remain stationary until the operator explicitly submits a new goal.	Baseline candidate	Test

5.5 Recovery

ID	Shall statement	Status	Verify
REC-001	Automatic recovery shall not by itself fail a trial; a completed trial satisfying all criteria shall be recorded as Pass - recovery used.	Baseline candidate	Test and inspection
REC-002	Following loss of a feasible path, the system shall attempt normal replanning for no more than 10 seconds before escalating recovery.	Provisional target	Test
REC-003	No more than one controlled local-costmap clear and repopulation shall be permitted in one recovery sequence.	Provisional target	Test
REC-004	The initial V1 recovery configuration shall not use backup or repeated spin behaviors.	Baseline candidate	Inspection and test
REC-005	The complete recovery sequence shall not exceed 30 seconds.	Provisional target	Test
REC-006	Exhausted recovery shall command zero motion, report failure, return stationary to Ready, and not restart the failed goal automatically.	Baseline candidate	Test

5.6 Live obstacles

ID	Shall statement	Status	Verify
OBS-001	A newly placed stationary obstacle observed by LiDAR shall appear in the active obstacle representation within 0.50 seconds of its first valid detection.	Provisional target	Test
OBS-002	A removed temporary obstacle shall clear from the active obstacle representation within three seconds when unobstructed LiDAR observations confirm free space.	Provisional target	Test
OBS-003	The local planner shall not command a path through a currently observed lethal obstacle even when that space is free in the saved map.	Baseline candidate	Test
OBS-004	TB-A, TB-B, and TB-C evidence shall include the relevant LiDAR, costmap, path, velocity-command, and mission-state records.	Baseline candidate	Inspection
OBS-005	The moving-cylinder stretch test shall evaluate whether changing obstacle position produces a predictable stop or avoidance response without contact.	Stretch	Exploratory test

5.7 Safety

ID	Shall statement	Status	Verify
SAFE-001	The robot shall register zero contact with walls, permanent obstacles, temporary obstacles, and controlled moving test objects during the frozen acceptance campaign.	Baseline candidate	Test
SAFE-002	The robot shall maintain at least 0.20 m separation between its collision geometry and obstacles during controlled mapping and navigation.	Provisional target	Test and analysis
SAFE-003	A temporary obstacle introduced ahead of a moving robot shall initially be at least 1.0 m from the front of its collision geometry.	Baseline candidate	Inspection and test
SAFE-004	A velocity command shall be considered stale when no valid replacement is received for 0.50 seconds.	Baseline candidate	Test
SAFE-005	Following watchdog expiry, the base controller shall issue or hold zero velocity within 0.10 seconds.	Provisional target	Test
SAFE-006	Within one second after watchdog expiry, measured linear speed shall be no greater than 0.01 m/s and angular speed no greater than 0.02 rad/s.	Provisional target	Test
SAFE-007	Motion after watchdog expiry shall not exceed 0.20 m before the stopped condition is achieved.	Provisional target	Test and analysis
SAFE-008	Stale or missing LiDAR, odometry, or required TF data shall inhibit new motion and stop active autonomous motion.	Baseline candidate	Fault-injection test
SAFE-009	A detected collision shall stop motion, fail the trial, preserve evidence, and inhibit further acceptance testing until reviewed.	Baseline candidate	Test
SAFE-010	Manual Safety Stop shall cancel the active goal, command zero velocity, latch motion inhibition, and require explicit operator reset followed by readiness checks.	Baseline candidate	Test
SAFE-011	Activation of the simulated Emergency Stop input shall bypass normal mission sequencing and initiate motion inhibition within 0.10 seconds.	Provisional target	Test
SAFE-012	Emergency Stop shall cancel the active goal, command zero, block subsequent motion commands, and remain latched until explicit reset.	Baseline candidate	Test

ID	Shall statement	Status	Verify
SAFE-013	Resetting Emergency Stop shall not produce motion and shall require the complete readiness sequence before a new goal can be accepted.	Baseline candidate	Test
SAFE-014	Diagnostics and evidence collection shall remain active when practical while simulated Emergency Stop is latched; orderly shutdown may follow only after motion inhibition is verified.	Baseline candidate	Test and inspection

5.8 Performance

ID	Shall statement	Status	Verify
PERF-001	The physics simulation shall use a fixed 60 Hz step during controlled trials.	Provisional target	Inspection and test
PERF-002	Mean simulation real-time factor shall be at least 0.95 during a valid trial.	Provisional target	Test and analysis
PERF-003	Real-time factor shall not remain below 0.90 for longer than two continuous seconds.	Provisional target	Test and analysis
PERF-004	LiDAR output shall maintain at least 9 Hz for the planned nominal 10 Hz configuration.	Provisional target	Test
PERF-005	Odometry and the required dynamic base TF chain shall each maintain at least 27 Hz for their planned nominal 30 Hz configuration.	Provisional target	Test
PERF-006	Required LiDAR, odometry, and dynamic TF data used by navigation shall not be older than 0.25 seconds.	Provisional target	Test
PERF-007	Command latency through the simulated robot interface shall not exceed 100 ms.	Provisional target	Test
PERF-008	The system shall operate for 30 continuous minutes without an unexplained crash, accumulating fault, or material timing degradation.	Baseline candidate	Test

5.9 Diagnostics

ID	Shall statement	Status	Verify
DIAG-001	The mission manager shall expose at least Preparing, Ready, Navigating, Recovering, Succeeded, Failed, Canceled, Manual Stop, Emergency Stop, and Inhibited states.	Baseline candidate	Test and inspection
DIAG-002	Startup and nominal trials shall contain no safety-critical alarm, unresolved functional error, or non-approved recurring warning affecting required operation.	Baseline candidate	Test and inspection
DIAG-003	Every rejected, canceled, timed-out, or failed mission shall record an outcome and actionable reason.	Baseline candidate	Test
DIAG-004	Expected automatic-recovery events shall be distinguishable from system-health faults.	Baseline candidate	Test and inspection

5.10 Evidence

ID	Shall statement	Status	Verify
EVID-001	Every controlled trial shall record requirement and test IDs, date and time, operator, scenario, start, goal, and result classification.	Baseline candidate	Inspection
EVID-002	Every controlled trial shall identify the repository revision and exact world, robot, map, launch profile, mission, and parameter versions used.	Baseline candidate	Inspection
EVID-003	Every controlled trial shall record completion time, final pose error, final mission state, collision status, interventions, recoveries, and applicable warnings or faults.	Baseline candidate	Inspection
EVID-004	Performance evidence shall record topic rates, data age, latency, real-time factor, resource use, CPU and GPU temperature, and detected thermal throttling.	Monitoring only	Inspection
EVID-005	Failed and invalid trials shall be retained and shall not be deleted or relabeled to improve reported performance.	Baseline candidate	Inspection
EVID-006	Final pose accuracy shall be measured using simulator ground truth independently from the localization estimate being evaluated.	Baseline candidate	Test and analysis
EVID-007	Every frozen requirement shall have a verification method, planned test or inspection, result, and supporting-evidence reference.	Baseline candidate	Inspection
EVID-008	Verification rigor shall be assigned from requirement risk and variability: high-risk motion responses require 10 consecutive passes, medium-risk variable behavior requires five repetitions, and deterministic low-risk items may use one test or inspection.	Baseline candidate	Inspection and analysis
EVID-009	Repeated tests shall report attempts, passes, failures, invalid trials, observed range, mean and standard deviation when meaningful, and a confidence interval for proportions.	Baseline candidate	Inspection and analysis
EVID-010	A failed repetition of a high-risk stop or motion-inhibition test shall remain in the evidence and shall reset the consecutive-pass sequence.	Baseline candidate	Inspection
EVID-011	Release-level reproducibility shall be demonstrated from the controlled setup procedure without undocumented configuration changes; independent execution is preferred and any absence of independent reproduction shall be recorded.	Baseline candidate	Demonstration

5.11 Configuration

ID	Shall statement	Status	Verify
CFG-001	Code, world, robot, maps, missions, parameters, launchers, and test procedures shall be version controlled or uniquely version identified.	Baseline candidate	Inspection
CFG-002	The acceptance configuration shall be frozen before the formal campaign and shall not change between counted trials.	Baseline candidate	Inspection
CFG-003	A change to a provisional value shall include measured justification, impact review, and a baseline revision before affected acceptance testing.	Baseline candidate	Inspection and analysis
CFG-004	A trial executed with an unidentified or incorrect configuration shall be classified as invalid.	Baseline candidate	Inspection

5.12 Acceptance

ID	Shall statement	Status	Verify
ACC-001	The nominal campaign shall contain 10 frozen start-goal scenarios, each performed three times, for 30 counted trials.	Baseline candidate	Inspection and test
ACC-002	At least 27 of the 30 nominal trials shall satisfy every successful-trial criterion.	Baseline candidate	Analysis
ACC-003	Each individual nominal scenario shall succeed in at least two of its three repetitions.	Baseline candidate	Analysis
ACC-004	The campaign shall include three clean cold-start and localization tests distributed across S1, S2, and S3.	Baseline candidate	Test
ACC-005	TB-A, TB-B, TB-C, localization loss, LiDAR loss, TF loss, command loss, cancellation, Manual Safety Stop, Emergency Stop, and shutdown shall be evaluated separately from the nominal success percentage.	Baseline candidate	Test and inspection
ACC-006	An invalid trial shall be retained, excluded from the pass-rate denominator, and repeated using the correct frozen procedure.	Baseline candidate	Inspection and analysis
ACC-007	A collision, uncontrolled-motion event, or failure to stop shall block release until investigated and formally dispositioned.	Baseline candidate	Inspection and analysis

5.13 Hardware transition

ID	Shall statement	Status	Verify
TRN-001	Before physical autonomous testing, the confirmed robot shall have an independently validated emergency-stop or motor-disable mechanism appropriate to its hardware; the V1 simulated Emergency Stop shall not be represented as physical safety validation.	Baseline candidate	Inspection and staged test

05 / TRACEABILITY AND VERIFICATION

Plan now; detail tests during implementation

6. Traceability rule

Every requirement is linked to its ConOps or Step 2 source, intended design allocation, verification method, planned test identifier, and risk-based verification rigor in the companion traceability CSV. Test procedures are authored when the feature is implemented, avoiding speculative documentation before interfaces exist.

6.1 Verification rigor

Class	Minimum evidence
High-risk motion, stop, or inhibition	10 consecutive successful repetitions per applicable condition, with zero observed failures; a failure remains in evidence and resets the consecutive-pass sequence.
Medium-risk variable behavior	Five controlled repetitions unless the nominal campaign already provides greater coverage.
Low-risk deterministic behavior	Automated test, one controlled demonstration, analysis, or inspection appropriate to the requirement.
Nominal navigation	Ten frozen scenarios repeated three times; report overall and per-scenario outcomes.
Repeated-result statistics	Attempts, passes, failures, invalid trials, range, mean/standard deviation where meaningful, and a confidence interval for proportions.

Statistical interpretation. Thirty nominal trials provide useful V1 engineering evidence, not production-reliability proof. A result of 27/30 is a 90 percent observed success rate but has a wide approximate 95 percent Wilson interval; the report must state the evidence without claiming universal 90 percent reliability.

7. Implementation allocations

The allocations deliberately remain component-level until Step 5 freezes nodes, topics, TF ownership, QoS, and command routing. The principal allocations are the launch/readiness manager, mode/command arbiter, SLAM and map QA, AMCL, Nav2 planner/controller/costmaps, mission manager, base watchdog, safety supervisor, diagnostics manager, evidence collector, performance monitor, configuration manager, and test harness.

06 / RISK REGISTER

Risks that change what we build or test

8. Risk method

The register is intentionally lightweight. Priority combines consequence and plausible occurrence for this V1. Each risk has an owner, observable trigger, mitigation, and planned verification. Detailed numerical risk scoring is unnecessary unless later decisions require it.

ID	Risk	Priority	Trigger	Primary mitigation / verification
R-01	Uncontrolled motion	High	Unexpected nonzero command or motion outside an authorized state	Single command owner, watchdog, Manual Safety Stop, Emergency Stop, state interlocks; TC-CMD-OWN-01; TC-WATCHDOG-01..04; TC-ESTOP-01..04
R-02	Collision or inadequate clearance	High	Clearance below 0.20 m or any contact	Validate footprint and geometry; conservative speeds; collision and clearance monitors; TC-CLEARANCE-01; TC-COLLISION-01..02

ID	Risk	Priority	Trigger	Primary mitigation / verification
R-03	Localization loss or false confidence	High	Covariance or pose freshness outside limits; implausible jump	Readiness gate, asymmetric world, health monitoring, motion inhibition; TC-LOC-READY-01; TC-LOC-LOSS-01
R-04	Stale or missing obstacle data	High	Data age above 0.25 s or missed rate budget	Freshness monitor, stop on loss, fault injection; TC-DATA-LOSS-01; TC-PERF-AGE-01
R-05	Incorrect or distorted map	High	Map accuracy or topology test fails	Map QA metrics, visual overlay, clean-restart test; TC-MAP-ACC-01..02; TC-MAP-TOPO-01
R-06	Temporary obstacle not marked or cleared	High	Marking or clearing latency exceeds target	Dedicated TB tests; retain scan and costmap evidence; TC-OBS-MARK-01; TC-OBS-CLEAR-01
R-07	Simulation slowdown	Medium	RTF or data-rate budget violated	Lightweight world, benchmarked launcher, performance logging; TC-PERF-RTF-01..02; TC-SOAK-01
R-08	Thermal throttling	Medium	Temperature trend and clock/utilization evidence indicate throttling	Monitor temperatures, reduce rendering load, stop invalid runs; TC-EVID-PERF-01; TC-SOAK-01
R-09	Incorrect robot geometry or dynamics	High	Professor or inspection confirms mismatch	Step 3 interface record; parameterize footprint, wheels, limits, and frames; Step 3 review; TC-NAV-SPEED-01
R-10	Software-version incompatibility	Medium	Package or API incompatibility appears	Freeze versions and installation procedure; clean-start smoke test; Step 4 environment test
R-11	Unrepeatable evidence	Medium	Trial lacks configuration identity or evidence	Named scenarios, automatic metadata, immutable failure retention; TC-EVID-META-01; TC-REPRO-01
R-12	Recovery loop or unsafe recovery	High	Recovery exceeds 30 s or repeats prohibited action	Bounded recovery, no backup/repeated spin, failure to Ready; TC-REC-ACT-01; TC-REC-LIMIT-01
R-13	Emergency Stop misunderstood as certified safety	High	Documentation or demo makes physical-safety claim	Explicit V1 limitation; separate V2 hardware gate; TC-V2-ESTOP-01; release review
R-14	Documentation overhead delays learning	Medium	Documentation expands without enabling a decision, test, or reproduction	One controlled baseline, generated CSVs, test details written during implementation; Step review
R-15	Single-developer knowledge concentration	Medium	Another user cannot execute smoke test	Lean operator instructions and release reproducibility test; TC-REPRO-01

9. Development versus acceptance baseline

Version 0.9 is the development baseline. Provisional targets may change during integration only when the change is supported by measurement, documented with rationale, reviewed for requirement/test impact, and incorporated before formal acceptance.

The Version 1.0 acceptance baseline will freeze the exact code revision, world, robot asset, map, scenario catalog, parameters, launcher, tolerances, timeouts, data-age limits, stop responses, and test procedures. Counted acceptance trials do not continue across a baseline-changing modification.

10. Professor and hardware information

Confirmed physical-robot facts are controlled in Step 3's physical interface record. This Step 2 baseline changes only when a confirmed physical constraint materially affects a V1 requirement or the V2 transition gate. Robot model, footprint, wheels, LiDAR/IMU, frames, topics, watchdog, onboard compute, ROS version, and existing Nav2 configuration are not silently inserted here as unverified facts.

08 / STEP 2 COMPLETION AND HANDOFF

Start building after a short review

11. Completion checklist

- ☐ Review the provisional targets and accept them as initial implementation targets.
- ☐ Incorporate any confirmed professor response that changes a V1 constraint; route remaining hardware facts to Step 3.
- ☐ Confirm every requirement has a design allocation and planned verification in the traceability CSV.
- ☐ Confirm every major risk has an owner, trigger, mitigation, and verification.
- ☐ Sign or record owner approval of Version 0.9 as the development baseline.
- ☐ Begin Steps 3–6 and implementation; do not wait for benchmark-dependent values to become final.

Step 2 exit. The project may proceed when every requirement is measurable or explicitly provisional, every major risk has a planned control, and no open question prevents safe simulation implementation. Version 1.0 is required before the formal acceptance campaign, not before coding begins.

12. Companion controlled files

File	Purpose
V1 Requirements Register v0.9 CSV	Sortable atomic requirements and statuses.
V1 Traceability Matrix v0.9 CSV	Source, design allocation, verification, planned test, risk class, result, and evidence link.
V1 Risk Register v0.9 CSV	Full cause, effect, priority, owner, trigger, mitigation, verification, and status fields.

13. References

1. *V1 Concept of Operations and Scope Baseline*, Version 0.9, Jan Cuesta, 23 September 2026.
2. *V1 Robot Development and Validation Plan*, Version 2.0, Jan Cuesta, 21 September 2026.
3. *Autonomous Semantic Navigation Robot – Source of Truth*, current controlled revision.
4. Nav2 documentation: mapping/localization, costmaps, obstacle layer, and action feedback.
5. ISO 13850:2015, *Safety of machinery – Emergency stop function – Principles for design*, used as terminology guidance only; no certification claim is made.